

Coach Attack Explorer

Interactive Visualisation Report

Goal build-up replay and tournament style comparison using StatsBomb event data

Field	Content
Project	Coach Attack Explorer
Module	Interactive Visualisation
Dataset	StatsBomb open event data, transformed into local goal build-up CSV files
Target users	Football coaches, football fans, beginner analysts
Date	27 May 2026

This report is written for the final IVI submission and documents the dashboard concept, data preparation, interaction design, evaluation setup and limitations.

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1. Introduction and motivation

The Coach Attack Explorer is an interactive dashboard for analysing how football teams create goals in major tournaments. Instead of focusing only on final scores or general statistics, the dashboard looks at the build-up before goals. This makes the analysis closer to real coaching questions, because coaches often want to understand how an attack developed before the final shot.

The main research question is: How do teams create goals in major tournaments, and how does their attacking style compare with their tournament outcome?

The project is designed as an exploratory visualisation. It does not try to predict winners or prove that one style is always better. The goal is to help users inspect concrete examples, compare teams and create practical tactical interpretations.

2. Use case and target users

The intended user is a football coach, a football fan with tactical interest or a beginner analyst. The dashboard should be understandable without advanced data-science knowledge. The user should be able to move from a concrete goal replay to a broader tournament comparison.

The dashboard supports the following workflow:

- Select a tournament from the prepared tournament list.
- Select one team and one goal example.
- Replay the goal attack step by step on the pitch.
- Inspect tournament-level goal patterns.
- Compare the selected team with other teams in the same tournament.

The practical coaching use case is that the user can take a professional goal example and discuss how the attack was created. For example, a coach can show whether a goal came from a quick direct attack, a medium build-up or a longer patient sequence.

3. Data source and preprocessing

The project uses StatsBomb open event data. The raw data contains ball actions such as passes, carries and shots. For the dashboard, these events were transformed into goal build-up sequences.

The final dashboard reads local processed files instead of downloading data during normal use. This decision was made because full tournament loading was too slow and unreliable during development and presentation. The local dataset makes the dashboard faster, more stable and reproducible.

File	Purpose
data/raw/selected_tournaments.csv	List of tournaments that can be selected in the dashboard.

data/processed/goals_df.csv	One row per analysed goal build-up with team, scorer, opponent, passes, duration and style.
data/processed/build_up_events_df.csv	One row per event in the selected attack sequence. Used for the pitch replay.
data/processed/team_efficiency_df.csv	Team-level summary values used for ranking and comparison.

The build-up style is classified with a simple rule based on completed passes before the goal. This keeps the interpretation clear for the target users. The categories are quick attack, medium build-up and long build-up.

4. Dashboard design and interaction concept

The final dashboard is structured around three main views. The first view is the Attack Replay. It shows a selected goal sequence on a football pitch and allows users to step through each event. The pitch is divided into build-up, progression and final-third zones to support tactical interpretation.

The second view is Tournament goal patterns. It provides a tournament-level overview with KPI cards, a build-up type distribution and a passes-versus-duration chart. These views answer whether the selected tournament contains mostly quick attacks, medium build-ups or longer goal sequences.

The third view is Team style and tournament finish. It contains a team style map and a ranking table. The x-axis in the style map represents the average completed passes before goals. Teams on the left score more directly, while teams on the right use longer build-ups. The selected team is highlighted. The ranking table connects directness with tournament finish.

The interaction design follows core information visualisation principles: overview first, filter, details on demand. Tournament selection changes all available teams and goals. Team and goal selection update the replay and comparison views. The replay then provides detailed inspection of one example.

5. Evaluation setup and example result structure

The evaluation is designed as a small formative usability test with two to three participants. The goal is not statistical proof, but checking whether the dashboard supports the intended workflow. Participants should represent realistic users such as football fans, students, amateur coaches or beginner analysts.

The evaluation uses five tasks. These tasks cover tournament selection, team selection, goal replay, interpretation of goal patterns and comparison with the tournament ranking.

Task	Goal of the task
1	Select a tournament and confirm that goal build-ups are loaded.
2	Select one team and one goal example.
3	Replay the attack and describe how the goal was created.
4	Use the overview charts to identify the most common build-up type.
5	Use the style map and ranking table to compare the selected team with the tournament.

For each participant, task completion, comments, main problems and SUS answers should be recorded. The CSV template in the evaluation folder is prepared for this. The example rows below show the expected structure and should be replaced by real participant results before final submission if required by the module.

Participant	Profile	Completion	Main feedback	Main problem
P1	Football fan	5/5	Replay was easy to understand and pitch zones helped.	Directness rank needed a short explanation.
P2	Student	4.5/5	Workflow was clear after trying it once.	Team style map was visually busy when many labels overlapped.
P3	Amateur coach	5/5	Replay was useful for coaching because it showed the attack development.	Build-up category thresholds needed explanation.

The intended analysis is to calculate task completion and the average SUS score. Qualitative feedback should be grouped into themes such as replay usability, clarity of ranking, label clutter and interpretation of build-up categories.

6. Design iteration

The dashboard changed significantly during development. The first versions had loading problems because full tournament data was loaded too slowly. This was solved by creating a local static dataset that is loaded directly by the app.

The second issue was layout complexity. Earlier versions contained additional information blocks that did not support the main task. These were removed to keep the replay and tournament comparison central.

The third issue was the comparison view. A single-team plot was not useful enough. The final version shows all teams in the selected tournament and highlights the selected team. This makes the comparison much stronger.

The fourth improvement was the ranking table. A pure directness ranking only showed style, but not tournament outcome. The final table therefore also includes tournament finish. This supports the main use case because users can compare how directly a team scores with how far it reached.

7. Limitations

The dashboard has several limitations that must be considered when interpreting the results.

- Only goal build-ups are analysed, not all attacks in a match.
- The dashboard is based on ball event data. Off-ball runs and defensive movements are not fully visible.
- The build-up style classification is simplified and based mainly on completed passes and duration.
- The ranking describes how teams scored their analysed goals. It is not a complete measurement of overall team quality.
- The dashboard does not prove that direct or patient play causes tournament success.

8. Conclusion

The Coach Attack Explorer provides a focused and practical interactive visualisation for football goal build-up analysis. Its main strength is the connection between a concrete attack replay and tournament-level comparison. Users can inspect how a goal was created and then compare the selected team against the rest of the tournament.

The project demonstrates several important IVI principles: linked views, filtering, overview and detail, interaction, visual comparison and evaluation planning. The dashboard is not a prediction system, but it gives users a clear way to explore how teams create goals and how their attacking style differs within a tournament.

Appendix: Evaluation tasks and SUS scoring

The evaluation tasks are documented in `IVI/evaluation/evaluation_tasks.md`. The SUS questionnaire is documented in `IVI/evaluation/sus_questionnaire.md`. For SUS scoring, odd-numbered answers are converted by subtracting 1. Even-numbered answers are converted by calculating 5 minus the response. The adjusted values are summed and multiplied by 2.5.

Before final submission, the example evaluation values should be replaced with real observations if the module requires measured user results.